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Gartner Identifies the Top Strategic Technology Trends for 2026

ORLANDO, Fla., October 20, 2025

Analysts Explore Industry Trends at Gartner IT Symposium/Xpo 2025, October 20-23

Gartner, Inc., a business and technology insights company, today announced its list of **top strategic technology trends** (https://www.gartner.com/en/articles/top-technology-trends-2026?utm_medium=press-release&utm_campaign=RM_GB_2025_ITCIO_NPP_PR1_26TTTPILLARPR) that organizations need to explore in 2026. Analysts presented their findings during **Gartner IT Symposium/Xpo** (<https://www.gartner.com/en/conferences/na/symposium-us>), taking place here through Thursday.



Gartner analysts Tori Paulman and Gene Alvarez on stage at Gartner IT Symposium/Xpo

“Technology leaders face a pivotal year in 2026, where disruption, innovation, and risk are expanding at unprecedented speed,” said **Gene Alvarez**

(<https://www.gartner.com/en/experts/gene-alvarez>), Distinguished VP Analyst at Gartner. “The top strategic technology trends identified for 2026 are tightly interwoven and reflect the realities of an AI-powered, hyperconnected world where organizations must drive responsible innovation, operational excellence, and digital trust.”

“These trends represent more than technology shifts; they are catalysts for business transformation,” said **Tori Paulman** (<https://www.gartner.com/en/experts/tori-paulman>), VP Analyst at Gartner. “What feels different this year is the pace. We’ve seen more innovations emerge in a single year than ever before. Because the next wave of innovation isn’t years away, organizations that act now will not only weather volatility but shape their industries for decades to come.”

The **top strategic technology trends for 2026**

([https://www.gartner.com/en/articles/top-technology-trends-2026?](https://www.gartner.com/en/articles/top-technology-trends-2026?utm_medium=press-release&utm_campaign=RM_GB_2025_ITCIO_NPP_PR1_26TTTPILLARPR)

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[release&utm_campaign=RM_GB_2025_ITCIO_NPP_PR1_26TTTPILLARPR](https://www.gartner.com/en/articles/top-technology-trends-2026?utm_medium=press-release&utm_campaign=RM_GB_2025_ITCIO_NPP_PR1_26TTTPILLARPR)) are:

AI Super Computing Platform

AI supercomputing platforms integrate CPUs, GPUs, AI ASICs, neuromorphic and alternative computing paradigms, enabling organizations to orchestrate complex workloads while unlocking new levels of performance, efficiency and innovation. These systems combine powerful processors, massive memory, specialized hardware, and orchestration software to tackle data-intensive workloads in areas like machine learning, simulation, and analytics.

By 2028, Gartner predicts that over 40% of leading enterprises will have adopted hybrid computing paradigm architectures into critical business workflows, up from the current 8%.

“This capability is already driving innovation across a diverse range of industries,” said Paulman. “For example, in healthcare and biotech, companies are modeling new drugs in weeks instead of years. In financial services, organizations are simulating global markets to reduce portfolio risk, while utility providers are modeling extreme weather to optimize grid performance.”

Multiagent Systems

Multiagent systems (MAS) are collections of AI agents that interact to achieve individual or shared complex goals. Agents may be delivered in a single environment or developed and deployed independently across distributed environments.

“Adopting multiagent systems gives organizations a practical way to automate complex business processes, upskill teams, and create new ways for people and AI agents to work together,” said Alvarez. “Modular, specialized agents can boost efficiency, speed up delivery, and reduce risk by reusing proven solutions across workflows. This approach also makes it easier to scale operations and adapt quickly to changing needs.”

Domain-Specific Language Models (DSLMS)

CIOs and CEOs are demanding more business value from AI, but generic large language models (LLMs) often fall short for specialized tasks. **Domain-specific language models** (<https://www.gartner.com/en/newsroom/press-releases/2025-07-10-gartner-forecasts-worldwide-end-user-spending-on-generative-ai-models-to-total-us-dollars-14-billion-in-2025>) (DSLMS) fill this gap with higher accuracy, lower costs, and better compliance. DSLMs are language models trained or fine-tuned on specialized data for a particular industry, function, or process. Unlike general-purpose models, DSLMs deliver higher accuracy, reliability, and compliance for targeted business needs.

By 2028, Gartner predicts that over half of the **GenAI** (https://www.gartner.com/en/industries/high-tech/trends/top-genai-opportunities-for-tech-products?utm_source=press-release&utm_medium=promotion&utm_campaign=RM_GB_2025_HTGBLPT_NPP_PR1_GENAIFORPRODUCT&utm_term=models) used by enterprises will be domain-specific.

“Context is emerging as one of the most critical differentiators for successful **agent** (https://www.gartner.com/en/doc/best-practices-preparing-for-ai-agent-services-opportunities?utm_campaign=RM_GB_2025_HTTPSL_NPP_PR1_AGENTICAIQ325) deployments,” said Paulman. “AI agents unpinned by DSLMs can interpret industry-specific context to make sound decisions even in unfamiliar scenarios, excelling in accuracy, explainability and sound decision-making.”

AI Security Platforms

AI security platforms provide a unified way to secure third-party and custom-built AI applications. They centralize visibility, enforce usage policies, and protect against AI-specific risks such as prompt injection, data leakage, and rogue agent actions. These platforms help CIOs enforce use policies, monitor AI activity, and apply consistent guardrails across AI.

By 2028, Gartner predicts that over 50% of enterprises will use **AI security** (https://www.gartner.com/en/cybersecurity/topics/cybersecurity-and-ai?utm_medium=press-release&utm_campaign=RM_GB_YOY_ITSEC_NPP_PR1_PILCYBERAI) platforms to protect their AI investments.

AI-Native Development Platforms

AI-native development platforms use **GenAI** (https://www.gartner.com/en/doc/779868-emerging-tech-top-use-cases-for-generative-ai?utm_campaign=RM_GB_2024_HT_NPP_PR1_GENA) to create software faster and easier than was previously possible. Software engineers embedded in the business, acting as “forward-deployed engineers,” can use these platforms to work together with domain experts to develop applications. Organizations can have tiny teams of people paired with AI to create more applications with the same level of developers they have today. Leading organizations are creating tiny platform teams to allow non-technical domain experts to produce software themselves, with security and governance guardrails in place.

Gartner predicts that by 2030, AI-native development platforms will result in 80% of organizations evolving large software engineering teams into smaller, more nimble teams augmented by AI.

Confidential Computing

Confidential computing changes how organizations handle sensitive data. By isolating workloads inside hardware-based trusted execution environments (TEEs), it keeps content and workloads private even from infrastructure owners, cloud providers, or anyone with physical access to the hardware. This is especially valuable for regulated industries and global operations facing geopolitical and compliance risks and for cross-competitor collaboration.

By 2029, Gartner predicts more than 75% of operations processed in untrusted infrastructure will be secured in-use by confidential computing.

Physical AI

Physical AI brings intelligence into the real world by powering machines and devices that sense, decide, and act, such as robots, drones, and smart equipment. It brings measurable gains in industries where automation, adaptability, and safety are priorities.

As adoption grows, organizations need new skills that bridge IT, operations, and engineering. This shift creates opportunities for upskilling and collaboration but may also raise job concerns and require careful change management.

Preemptive Cybersecurity

Preemptive cybersecurity (<https://www.gartner.com/en/newsroom/press-releases/2025-09-18-gartner-says-that-in-the-age-of-genai-preemptive-capabilities-not-detection-and-response-are-the-future-of-cybersecurity>) is trending as organizations face an exponential rise in threats targeting networks, data, and connected systems. Gartner forecasts that by 2030, preemptive solutions will account for half of all security spending, as CIOs shift from reactive defense to proactive protection.

“Preemptive cybersecurity is about acting before attackers strike using AI-powered SecOps, programmatic denial and deception,” said Paulman. “This is a world where prediction is protection.”

Digital Provenance

As organizations rely more on third-party software, open-source code, and AI-generated content, verifying digital provenance has become essential. Digital provenance refers to the ability to verify the origin, ownership, and integrity of software, data, media, and processes. New tools such as software bills of materials (SBOM), attestation databases, and digital watermarking offer organizations the means to validate and track digital assets across the supply chain.

Gartner predicts that by 2029, those who failed to adequately invest in digital provenance capabilities will be open to sanction risks potentially running into the billions of dollars.

Geopatiation

Geopatiation means moving company data and applications out of global public **clouds** (<https://www.gartner.com/en/infrastructure-and-it-operations-leaders/topics/cloud-computing>) and into local options such as sovereign clouds, regional cloud providers, or an organization's own data centers due to perceived geopolitical risk. Cloud sovereignty, once limited to banks and governments, now affects a wide range of organizations as global instability increases.

“Shifting workloads to providers with an increased sovereignty posture can help CIOs gain more control over data residency, compliance and governance,” said Alvarez. “This greater control may improve alignment with local regulations and build trust with customers who are concerned about data privacy or national interests.”

Gartner predicts that by 2030, more than 75% of European and Middle Eastern enterprises will geopatiate their virtual workloads into solutions designed to reduce geopolitical risk, up from less than 5% in 2025.

This year's top strategic technology trends highlight those trends that will drive significant disruption and opportunity for CIOs, IT and high tech leaders within the next five years. Gartner clients can read more in the Gartner Special Report **“Top Strategic Technology Trends for 2026.”** (<https://www.gartner.com/document-reader/document/7079798>)

About Gartner IT Symposium/Xpo

Gartner IT Symposium/Xpo (<https://www.gartner.com/en/conferences/hub/cio-conferences>) is the world's most important gathering for CIOs and other IT executives. CIOs and IT executives rely on these conferences to learn how to become agents of change in their organizations and harness AI for successful digital transformation. Follow news and updates from the conference on **X** (https://x.com/Gartner_inc) and **LinkedIn** (<https://www.linkedin.com/showcase/gartner-for-it-leaders>) using

([#GartnerSYM](https://twitter.com/search?q=%23GartnerSYM&src=typd), and on the **Gartner Newsroom** (<https://www.gartner.com/en/newsroom/topics/information-technology>).

Upcoming dates and locations for Gartner IT Symposium/Xpo include:

October 28-30, 2025 | Yokohama, Japan

(<https://www.gartner.com/jp/conferences/apac/symposium-japan>)

November 10-13, 2025 | Barcelona, Spain

(<https://www.gartner.com/en/conferences/emea/symposium-spain>)

November 17-19, 2025 | Kochi, India

(<https://www.gartner.com/en/conferences/apac/symposium-india>)

About Gartner AI Use Case Insights

Gartner AI Use Case Insights is an interactive tool that helps technology and business leaders efficiently discover, evaluate, and prioritize AI use cases to potentially pursue. Clients can search over 500 use cases (applications of AI in specific industries) and over 380 case studies (real world examples) based on industry, business function, and Gartner's assessment of potential business value. Clients can access the interactive tool at <https://tools.gartner.com/use-case-insights> (<https://tools.gartner.com/use-case-insights>).

About Gartner for High Tech Leaders and Providers

Gartner for High Tech Leaders and Providers equips tech leaders and their teams with role-based best practices, industry insights and strategic views into emerging trends and market changes to achieve their mission-critical priorities and build the successful organizations of tomorrow. Additional information is available at (<https://www.gartner.com/en/industries/high-tech>)www.gartner.com/en/industries/high-tech (<https://www.gartner.com/en/industries/high-tech>).

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Q&A | Apr 22, 2026

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Announcement | Apr 22, 2026

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